

## SUMMARY TABLE: GENERAL SCIENCE GRADE 10

### Sub-Strand 3.4: Magnetism and Electromagnetic Induction — Teacher Reference (Pre-filled)

| SUMMARY TABLE: GENERAL SCIENCE GRADE 10 |                                                                                                                                                                      |
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| <b>Sub-Strand</b>                       | Sub-Strand 3.4: Magnetism and Electromagnetic Induction                                                                                                              |
| <b>Driving Question</b>                 | DRIVING QUESTION: How does moving a wire through a magnetic field produce electricity — and how is this principle powering homes, farms and industries across Kenya? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                            | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Lesson 1</b><br><b>Anchoring Phenomenon &amp; Methods of Magnetisation</b> | A bicycle dynamo lights a small LED only when the wheel is spinning — the light goes out the moment the wheel stops. A hand-cranked torch behaves the same way. Inside the Kericho school generator, large copper coils rotate near magnets to power the entire dormitory — no battery or chemical reaction is involved. In our magnetisation experiments, the stroking method (20 unidirectional strokes) produced the strongest magnet on average, the electrical (solenoid) method produced a strong magnet that was demagnetised when disconnected, and the induction method produced the weakest magnet. | Soft iron can be magnetised by three methods: (1) STROKING — repeated unidirectional strokes with one pole of a bar magnet physically align the magnetic domains in the iron; (2) ELECTRICAL (solenoid) — passing an electric current through a coil wound around the iron creates a magnetic field that aligns the domains (this produces an electromagnet — temporary); (3) INDUCTION — placing the iron close to a strong magnet allows the external field to align domains without contact (produces the weakest magnet of the three; in soft iron this induced magnetism is temporary and is mostly lost once the magnet is removed). Magnetic domains are regions of aligned atomic magnets; an unmagnetised piece of iron has randomly oriented domains. A magnet is essential inside any generator or dynamo — without it, no electromagnetic induction can occur. | Why is movement essential for the dynamo to produce electricity? (To be fully explained in Lessons 3–4.) Why does faster spinning make the LED brighter? (To be fully explained in Lesson 4 — factors affecting induced e.m.f.) Why does the same spinning-coil principle that lights a bicycle dynamo also power Kenya's national grid? (To be fully explained in Lesson 5–6 — applications of electromagnetic induction.) Why is soft iron used as the core material in generators rather than steel? (Partially addressed today — soft iron is easily magnetised and demagnetised; full explanation of core function in Lesson 3.) |
| <b>Lesson 2</b><br><b>Demagnetisation &amp; Properties of Magnets:</b>        | Learners observed that a magnetised nail lost its ability to deflect a compass needle after being heated to red-hot over a jiko; a                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Learners learned that magnetism depends on the ordered alignment of magnetic domains; any process that supplies enough                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Learners explained that the Kericho student's bicycle dynamo light goes out when the wheel stops because stopping the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| <b>Why Stillness Kills the Current</b>                                                            | magnetised iron rod lost its deflection after 20 hammer strikes on an anvil block; and a magnetised rod showed zero galvanometer response after being slowly withdrawn through an AC solenoid — demonstrating three distinct pathways to demagnetisation.                                                                                                                                                                                                                                                                                       | energy (heat, mechanical shock) or repeatedly reverses the field direction (alternating current) randomises the domains and destroys magnetism. This domain disorder is analogous to why a stationary generator produces no current — without continuous rotation, the conditions for ordered induction are not maintained.                                                                                                                                                                                                                                                                                                                                                                                                                                       | wheel is functionally equivalent to demagnetisation — the ordered relationship between the magnetic field and the moving conductor collapses, just as ordered domains collapse when a magnet is heated or hammered. Engineers at Kenya Power protect generator magnets from heat and vibration for exactly this reason: domain disorder means power failure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lesson 3<br/>Magnetic Field Patterns:<br/>Mapping the Invisible Force</b>                      | Learners sprinkled iron filings over bar magnets and U-shaped magnets and used plotting compasses to trace the direction of magnetic field lines, producing two annotated field diagrams showing field line patterns, direction arrows (North to South outside the magnet), crowding of lines at poles, and the uniform parallel-line pattern between the poles of the U-shaped magnet.                                                                                                                                                         | Magnetic fields have direction (field lines run from the North pole to the South pole outside a magnet); field line density indicates field strength — lines are most crowded at the poles where the field is strongest; the U-shaped magnet produces a uniform, intense magnetic field in the region between its poles; this uniform field region is exactly where the copper coils of the Kericho school generator (and all generators) rotate, because the strong, constant field in that zone means the moving conductor cuts more field lines per second, which maximises the induced e.m.f.                                                                                                                                                                 | The Kericho boarding school generator works because the large copper coils rotate inside a housing that functions like a giant U-shaped magnet — the coils spin through the intense, uniform field between the poles. The bicycle dynamo that the student removed from the old bike uses the same principle on a smaller scale. Today's lesson makes the invisible field visible: where field lines are densest, the magnetic force is greatest, and it is in that dense-field zone that movement of a conductor will produce the largest induced e.m.f. — which is why faster spinning of the dynamo wheel produces a brighter LED light.                                                                                                                                                                                                 |
| <b>Lesson 4<br/>Introducing Induced E.M.F (Qualitative): What Makes the Galvanometer Deflect?</b> | Learners observed that a straight copper conductor held stationary between the poles of a U-shaped magnet produces zero galvanometer deflection; that moving the conductor parallel to field lines produces zero or negligible deflection; that moving the conductor vertically upward (cutting across field lines) produces a clear deflection; and that moving it at an angle produces a deflection of intermediate magnitude. Learners also observed that faster movement in scenario (c) produces a larger deflection than slower movement. | Learners learned that an electromotive force (e.m.f) is induced in a conductor only when there is relative motion between the conductor and the magnetic field, and specifically only when the conductor cuts across field lines rather than moving parallel to them. This is the principle of electromagnetic induction. The direction of deflection reverses when the conductor moves downward instead of upward, indicating that the direction of induced current depends on the direction of motion. Learners connected this principle to the Kericho school generator and the bicycle dynamo: in both cases, continuous rotational motion keeps the conductor cutting field lines, sustaining a continuous induced e.m.f and therefore a continuous current. | Learners can now explain why Akoth's bicycle dynamo LED lights up only when the wheel is spinning: rotation causes the conductor inside the dynamo to continuously cut through magnetic field lines, inducing an e.m.f that drives current through the LED. When the wheel stops, the conductor becomes stationary relative to the field, the cutting action ceases, the induced e.m.f drops to zero, and the LED goes out. The same principle applies to the Kericho school generator: the large copper coils must rotate continuously relative to the fixed magnets (or electromagnets) to sustain the induced e.m.f that powers the dormitory lights, water pumps, and kitchen equipment. No battery or chemical reaction is needed — mechanical motion is converted directly into electrical energy through electromagnetic induction. |
| <b>Lesson 5</b>                                                                                   | Learners observed galvanometer needle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Learners learned that four factors control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Learners explained that the student in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| <b>Factors Affecting the Magnitude of Induced E.M.F.</b>           | <p>deflections change in magnitude as they varied: (a) the speed of pushing a bar magnet into a coil — faster push → larger deflection; (b) the number of stacked magnets — two magnets → larger deflection than one; (c) the number of coil turns — 100-turn coil → larger deflection than 20-turn coil; (d) the angle of magnet entry — 90° (perpendicular) → maximum deflection; 0° (parallel) → zero deflection. Learners also observed that the galvanometer needle returned to zero immediately when movement stopped — reinforcing the Kericho dynamo observation that the LED goes out when the wheel stops.</p> | <p>the magnitude of induced e.m.f.: speed of conductor/magnet movement, strength of the magnetic field, number of turns of wire in the coil, and angle of movement through the field. They learned that all four factors share a single underlying principle — Faraday's Law — that the induced e.m.f. is proportional to the rate of change of magnetic flux through the coil. Learners also learned how to design a fair investigation by controlling variables, how to record results systematically in a table, how to plot and read a bar chart of electromagnetic data, and how to construct a Claim-Evidence-Reasoning explanation from experimental results.</p>              | <p>Kericho sees the LED go brighter when she cranks the dynamo faster because increasing speed increases the rate at which magnetic field lines are cut by the coil, increasing the induced e.m.f. They explained that Kenya's large power-station generators (e.g., Olkaria IV geothermal, Tana River hydroelectric) produce thousands of volts because engineers maximise all four factors simultaneously: rotating coils at high speed, using powerful electromagnets, winding thousands of turns of copper wire, and designing the geometry so conductors always cut field lines at 90°. Learners used the CER framework to replace their earlier predictions on the DQB with evidence-based statements, and revised their cumulative electromagnetic induction model to show all four magnitude-controlling factors with directional annotations.</p> |
| <b>Lesson 6 Applications of Electromagnetic Induction in Kenya</b> | <p>Learners used digital devices and print media to research six real-world applications of electromagnetic induction in Kenya: AC generators, transformers, the Olkaria Geothermal Plant (Naivasha), the Lake Turkana Wind Farm (Marsabit), electric bells and bicycle dynamos. Each expert group gathered evidence and presented findings to the class, with specific reference to how each application connects to experimental data from Lessons 4 and 5.</p>                                                                                                                                                        | <p>All six applications — from the bicycle dynamo on the Kericho student's shelf to the 310 MW Lake Turkana Wind Farm — operate on the same principle: relative motion between a conductor and a magnetic field induces an electromotive force. The magnitude of this e.m.f. is controlled by the same factors identified in the lab: speed of rotation, number of coil turns and strength of the magnetic field. Kenya Power's national grid depends entirely on this principle: step-up transformers at Olkaria raise voltage to 132 kV for transmission, and step-down transformers reduce it to 240 V at the point of use — both relying on changing magnetic flux in a coil.</p> | <p>The Kericho student's generator room observation is no longer puzzling: the large copper coils she sees rotating inside the metal housing are the same coil-in-a-magnetic-field setup she demonstrated in the lab — just scaled up enormously. The coils spin at high speed (maximising rate of change of flux), have thousands of turns (maximising induced e.m.f.) and are wound around strong electromagnet cores (maximising field strength). When the wheel of her bicycle dynamo stops, relative motion ceases, the magnetic flux through the coil no longer changes, and — by Faraday's law — the induced e.m.f. immediately drops to zero, which is why the LED goes out instantly. The same physics, operating at every scale from a bicycle wheel to a geothermal turbine, is what powers Kenya's homes, farms and industries.</p>            |
| <b>Lesson 7 Project: Design and Make a Simple Electric Bell</b>    | <p>Groups built a simple electric bell using an iron nail, copper wire, dry cell, thin metal strip (armature), wooden base and drawing pins. They observed that when the circuit was closed, the nail electromagnet attracted</p>                                                                                                                                                                                                                                                                                                                                                                                        | <p>The electric bell works because of controlled, repeated magnetisation and demagnetisation of soft iron. The iron nail becomes an electromagnet (magnetisation) when current flows through the coil; it</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>Movement (or current flow) is essential to electromagnetic action: when it stops, so does the electromagnetic effect — whether in a bicycle dynamo LED, a school electric bell, or the Kericho generator. The bell's</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|                                                                                                                                                     | <p>the metal strip; when the strip touched the nail, the circuit broke; the nail demagnetised; the strip sprang back; the circuit closed again — producing a repeated ringing sound. Groups that added more coil turns or increased current observed a louder, faster ring. Groups that held the armature against the nail observed that ringing stopped immediately — directly paralleling the Kericho student holding the bicycle dynamo still.</p> | <p>attracts the armature and breaks the circuit; without current the nail loses its magnetism (demagnetisation) and the armature springs back to close the circuit again. This make-and-break cycle is the same principle that governs the switching of polarity in a generator's rotating coil. The number of coil turns and the current strength (factors affecting the magnitude of induced e.m.f.) directly affect how strongly and quickly the cycle operates — connecting today's bell directly to the Kericho generator, which uses thousands of coil turns and powerful electromagnets to produce enough current for the entire dormitory.</p> | <p>electromagnet switching on and off is a miniature, visible demonstration of the same controlled magnetisation and demagnetisation that occurs continuously in industrial generators. This principle — underpinning the KICD key inquiry question 'How is the study of electromagnetic induction important in our daily life?' — powers school bell systems, water pumps on Rift Valley farms, fish-processing equipment in Kisumu, and the national grid that supplies Kenya's homes and industries.</p>                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Lesson 8</b><br/><b>Model Building, DQB</b><br/><b>Closure and Final</b><br/><b>Assessment: Answering</b><br/><b>the Driving Question</b></p> | <p>We observed that moving a copper conductor in a magnetic field causes a galvanometer to deflect, and that the deflection is larger when the conductor moves faster, the magnet is stronger, more coil turns are used, and the conductor moves perpendicular to the field. We also observed that stillness produces zero deflection — the conductor must be moving relative to the magnetic field for any current to be induced.</p>                | <p>We learned that electromagnetic induction is the process by which mechanical motion produces electrical energy without any battery or chemical reaction. We learned that the magnitude of the induced e.m.f. depends on four factors: speed of movement, strength of the magnetic field, number of turns of wire, and angle of the conductor relative to the field. We also learned that this single principle explains how the Kericho school generator, the bicycle dynamo, the Olkaria Geothermal Plant turbines and the Lake Turkana Wind Farm all produce electricity for Kenyan homes, farms and industries.</p>                              | <p>We can now fully explain why the Kericho student sees electricity produced from spinning coils with no battery: the mechanical rotation of the coil in the magnetic field continuously changes the flux through the coil, inducing an e.m.f. that drives current through the external circuit. The faster the coil spins, the greater the induced e.m.f. and the brighter the lights. The same principle explains why the bicycle dynamo LED goes out when the wheel stops — without relative motion between the conductor and the magnetic field, no e.m.f. is induced and no current flows. Our electric bell project from Lesson 7 showed that controlled magnetisation and demagnetisation of soft iron is the same switching mechanism used in generator coils, confirming that all sub-strand concepts are interconnected aspects of one fundamental electromagnetic phenomenon.</p> |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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